GERBANG BARITO REDD+ PROJECT

Project title	Gerbang Barito REDD+ Project
Project ID	5799
Crediting period	01-August-2023 to 31-July-2053
Original date of issue	03-December-2025
Most recent date of issue	03-December-2025
Version	1.0
VCS Standard Version	VCS v4.7
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1 PROJECT DETAILS

1.1 Summary Description of the Project

The Gerbang Barito REDD+ Project (hereinafter referred to as the GBRP), is a peat swamp forest ecosystem conservation, restoration, and livelihood development project under the United Nations' Reducing Emissions from Deforestation and Degradation (REDD+), located in the village forests of Batampang and Batilap within the administration of the Gerbang Barito Forest Management Unit (FMU) IX Protected Forest in South Barito District, Province of Central Kalimantan, Indonesia. The Project Area is 19,752 hectares, comprising 9,734 and 10,019 hectares for Batampang and Batilap Village Forests respectively. The two village forests are exclusive to and managed by the Village Forest Management Institution (LPHD or Lembaga Pengelola Hutan Desa) of each village (herein referred to as the project proponents) with regular supervision by the Gerbang Barito FMU. These peat swamp forests contain a specialized biodiversity that is adapted to the highly acidic and waterlogged area, which stores more carbon than other forest habitats in comparison to their small area.

The communities of Batampang and Batilap Villages consist of approximately 2,477 individuals in total, where their livelihoods are largely dependent on the ecosystem services of the forests. The growing population in the region is one of the biggest threats towards the peat swamp forest natural ecosystem's integrity. Many anthropogenic threats have manifested in the Project Area before 1990, including logging, wildlife hunting, and conventional fishing, all of which resulted in subsequent draining and drying of the peat, posing a risk of fires to the forests during dry seasons, with the addition of climate change exacerbating the risks. Meanwhile, the peat swamp forests are home to endangered umbrella and charismatic wildlife species such as Bornean orangutans and Proboscis monkeys. Increasing deforestation, occurrence of flooding, and unpredictable climate patterns have also threatened local river-based livelihoods and food security. The communities' economic vulnerabilities and limited infrastructure, combined with the ongoing environmental threats pose significant risks to the sustainability of the peat forests.

The project is expected to generate a climate benefit of 274,487 tCO2e annual average and total reductions and removals.

1.2 Audit History

Audit type	Period	Program	Validation/verification body name	Number of years
Validation/ verification	Validation Date: Expected to be initiated by 01-February-2026	VCS / CCB	A qualified VVB will be contacted for the Project's combined validation & verification prior to	2.42 years

	Initial Monitoring Period for Verification: 01-August-2023 - 31-December-2025		submitting the Project for Listing "Under Validation" on the Verra Registry.	

1.3 Sectoral Scope and Project Type

Sectoral scope	14 - AFOLU
AFOLU project category[1]	REDD+, Wetlands Restoration and Conservation (WRC)
Project activity type	• Avoiding Unplanned Deforestation and/or Degradation (AUDD): Activities that reduce net greenhouse gas ("GHG") emissions by stopping or reducing deforestation or degradation on forest lands that are not legally authorized and documented for conversion to other land use. • Conservation Intact Wetland (CIW): Activities that reduce GHG emissions by avoiding degradation and/or the conversion of wetlands that are intact or partially altered while still maintaining their natural functions. • Restoration of Wetlands Ecosystems (RWE): Activities that reduce GHG emissions by the restoration of wetlands that are drained by rewetting and returning them to their original natural functions.

1.4 Project Eligibility

1.4.1 General eligibility

The project type falls under the AFOLU (Agriculture, Forestry and Other Land Uses) sector (VCS 14), within the category of Reduced Emissions from Deforestation and Degradation (REDD) and Wetlands Restoration and Conservation (WRC) according to the VCS Standard v4.7. The project Avoids Unplanned Deforestation and Degradation (AUDD) and more specifically, it avoids unplanned mosaic deforestation. REDD AUDD project activities in the AFOLU sector (VCS Scope 14) are eligible under the VCS Program. Under WRC, the project will implement wetland conservation (CIW) or avoid degradation and conversion of wetlands and restore parts of the peat forest (RWE) due to drainage canals.

[1] *See Appendix 1 of the* VCS Standard

The methodology used for the Project is the VCS methodology VM0007 (REDD+ Methodology Framework) v1.8 of 4th June 2024, along with the relevant modules to quantify RWE and CIW in the allocated portions of the Project Area. This methodology quantifies greenhouse gas emission reductions generated on peat forest ecosystems from avoiding either planned or unplanned (or both) deforestation as initiated by a variety of agents and drivers.

The project is expected to meet all relevant deadlines, adhering to VCS Standard v4.7 of 16 April 2024.

- Prior to submitting the full VCS / CCB PD & MR to Verra, the project will firstly request the approval of the Ministry of § Environment, as required by the new Presidential Regulation No. 110 of 2025, which will take a minimum of five working days. After obtaining the approval, the project seeks to request public comment approval from Verra around January to February 2026, submitting the full VCS / CCB PD & MR to listing "under validation" on the Verra Registry and subsequently initiate joint validation and verification with the contracted VVB. The public comment period will also be discreetly announced to relevant stakeholders of the project.
- The project start date is stipulated in the same month as the first round of FPIC consultations with community members, i.e., August 2023. As an AFOLU project, the time range between the project start date and the project pipeline listing will be within three years from the start date (August 2023 to August 2026).
- The project strives to conform with the relevant laws and regulations as defined in Section 2.5.

1.4.2 AFOLU project eligibility

The AFOLU project categories are eligible according to VCS Standard v4.7, i.e., REDD and WRC. AFOLU REDD activities principally comprise of Avoided Unplanned Deforestation and Degradation (AUDD), while AFOLU WRC activities consist of CIW and RWE. The Project will permanently protect the remaining secondary swamp forest areas from unplanned conversion to other land cover/use and reduce net GHG emissions by reducing deforestation and deforestation. Through CIW, the Project will permanently protect the peatland soil areas from unplanned conversion, and from existing and future peatland drainage due to unplanned activities, usually for access and transport. Through RWE, the project will reduce GHG emissions through the restoration of existing drained and degraded peatlands. Rewetting is planned to improve groundwater table through canal backfilling and blocking of mapped drainage canals as well as revegetation to restore their original natural functions if required.

The GBRP area is mostly comprised of secondary peat swamp forest, in which the establishment of the Project Area corresponds to Gerbang Barito FMU's protected forest landscape that has remained as forests throughout the period of 2009-2020. Therefore, for more than ten years prior to the project start date, there was no clearing or conversion within the Project Area, and no native ecosystems have been converted, cleared, drained, or degraded to generate GHG credits. This evidence of unchanged forest cover for more than ten years is shown in Figure 1.1.

Figure 1.1 Two maps showing that the Project Area (red line) was peat swamp forest for the last 10 years from 2009 to 2020 and was not cleared or drained of existing natural ecosystems.

1.4.3 Transfer project eligibility

This section is not applicable, as this is not a transfer project.

1.5 Project Design

☒ Single location or installation

☐ Multiple locations or project activity instances (but not a grouped project)

☐ Grouped project

1.5.1 Grouped project design

This is not a grouped project. Therefore, this section is not relevant.

1.6 Project Proponent

Provide contact information for the project proponent(s). Copy and paste the table as needed.

Organization name	Lembaga Pengelola Hutan Desa (LPHD) Batampang
Contact person	Mr. Samitro
Title	Chairman
Address	Batampang Village, Dusun Hilir Sub-District, South Barito District, Central Kalimantan Province 73762
Telephone	(+62) 853-9854-0295
Email	lphddesabatampang@gmail.com

Organization name	Lembaga Pengelola Hutan Desa (LPHD) Batilap
Contact person	Mr. Jumani
Title	Chairman
Address	Batilap Village, Dusun Hilir Sub-District, South Barito District, Central Kalimantan Province 73762
Telephone	(+62) 821-4875-1397
Email	lphdbatilap@gmail.com

1.7 Other Entities Involved in the Project

Organization name	Wildlife Works
Role in the project	Project Developer
Contact person	Mr. Jeremy Freund
Title	Chief Climate Officer

Address	495 Miller Avenue, Suite 100, Mill Valley, CA 94941 USA
Telephone	+1 415 332 8081
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1.8 Ownership

The project ownership belongs to Village Forest Management Institution (LPHD) of Batampang and Batilap Village, community-led organizations who are granted the rights by the government under the MoEF Regulation No. 83/2016 to manage village forests as Protected Forest or *Hutan Lindung* (HL). Under the Social Forestry regulation, village forests or *Hutan Desa* in the Indonesian language, are state-owned forests granted to the village to be managed by the village apparatus (LPHD) for the community members' benefit. On 07 December 2017, the MoEF (now divided into two separate bodies, the Ministry of Environment and the Ministry of Forestry) legally granted these LPHDs a specific amount of forest area as village forest through an official decree of Village Forest Management Right (HPHD), after the village submitted its proposal to manage their forest to the MoEF and successfully underwent technical verification process. The decrees conform with VCS Standard's Article 1 of Section 3.7.1 and CCB G5.7. With both grants issued on 07 December 2017, the Batampang and Batilap Village Forest licenses validity period will end by 17 December 2052.

The following are the details of the granting of the official HPHD decrees.

a. Batampang Village Forest, as granted by Social Forestry Approval SK.6663/MENLHK-PSKL/PKPS/PSL.0/12/2017 for 9,668 ha of protected forests area in Batampang Village for 35 (thirty-five) years since 07 December 2017.

b. Batilap Village Forest, as granted by SK.6661/MENLHK-PSKL/PKPS/PSL.0/12/2017 for 9,951 ha of protected forests area in Batilap Village for 35 (thirty-five) years since 07 December 2017.

Technically, these forest areas are under the jurisdiction of the Gerbang Barito Forest Management Unit (FMU) Unit IX Protected Forest with a total size of 92,429 hectares based on the Ministry of Forestry Decree No. SK.2/Menhut-II/2012. The FMUs are legally obligated bodies that are established to carry out some of the operational and/or supporting technical or field tasks of the Central Kalimantan Provincial and South Barito District Forestry Service. Therefore, the management of the forests will be conducted by the LPHD under the supervision and support of the FMU and in coordination with the MoF, particularly with the Directorate of Social Forestry. In relation to village forests, FMUs are tasked with facilitating the formulation of forest village management activities, enhancing the capacity of the Village Forest Management Institutions (LPHDs), and assisting strategic partners in the management and utilization of village forests. On the other hand, the MoF and its provincial office administer social forestry affairs, including business groups that utilize forest products, land rehabilitation, and community empowerment.

The decision-making process in LPHD is through a forum and deliberations for consensus among all members with some involvement and intervention from the village government. LPHD as a community-led organization can appoint and manage community members in the village in relation to the utilization and protection of the village forest, such as harvesting NTFPs and patrolling the area from illegal loggers and overharvesting. This indirectly gives the communities the management rights over the Project Area, in which according to both approval decrees, the management rights over the protected forests include environmental services and carbon stock management activities. More details on the roles and responsibilities of the LPHD as an organization are provided in Section **Error! Reference source not found.**.

In accordance with the above, the ownership of the project is supported by numeral 4 - Section 3.7.1 of the VCS Standard requirements, which states:

"Project ownership arising by virtue of a statutory, property or contractual right in the land, vegetation or conservational or management process that generates GHG emission reductions and/or removals (where the project proponent has not been divested of such project ownership)."

Therefore, during the origination and implementation of the project, the carbon rights lie with the communities under the management of their LPHD. Under the supervision and support of Gerbang Barito FMU and the MoF, Batampang and Batilap LPHDs will be the project proponent of this project.

1.9 Project Start Date

Project start date	01 August 2023
Justification	The project start date conforms with VCS Standard v4.7 section 3.8. Before the start date, the project proponent had agreed upon a Memorandum of Understanding with Wildlife Works as the project partner, which shifted both LPHDs' management activities in the Project Area. Pre-FPIC activities were also done in the months prior, which led to the first round of FPIC consultation conducted at the same time as the start date. This start date conforms with the pipeline listing period of AFOLU projects within three years.

1.10 Project Crediting Period

Crediting period	☐ *Seven years, twice renewable* ☐ *Ten years, fixed* ☒ *Other (state the selected crediting period and justify how it conforms with the VCS Program requirements)*

	The project's first crediting period is 30 years (2023–2053), corresponding to the current forest village license validity period of 35 years. The village forest license can potentially be renewed, allowing a second crediting period of 35 years. This results in a total intended crediting duration and project longevity of 64 years. This exceeds Verra's minimum 40-year project longevity requirement as specified in VCS Standard v4.7.
Start and end date of first or fixed crediting period	01 August 2023 *to* 31 July 2053

1.11 Project Scale and Estimated GHG Emission Reductions or Removals

☒ < 300,000 tCO2e/year (project)

☐ ≥ 300,000 tCO2e/year (large project)

Calendar year of crediting period	Estimated GHG emission reductions or removals (tCO_2e)
01-August-2023 to 31-December-2023	48,422
01-January-2024 to 31-December-2024	116,213
01-January-2025 to 31-December-2025	130,342
01-January-2026 to 31-December-2026	144,471
01-January-2027 to 31-December-2027	158,600
01-January-2028 to 31-December-2028	172,729
01-January-2028 to 31-December-2029	186,858
01-January-2030 to 31-December-2030	198,460
01-January-2031 to 31-December-2031	207,744
01-January-2032 to 31-December-2032	217,028
01-January-2033 to 31-December-2033	226,312

01-January-2034 to 31-December-2034	235,596
01-January-2035 to 31-December-2035	244,880
01-January-2036 to 31-December-2036	254,164
01-January-2037 to 31-December-2037	263,448
01-January-2038 to 31-December-2038	272,732
01-January-2039 to 31-December-2039	282,016
01-January-2040 to 31-December-2040	291,300
01-January-2041 to 31-December-2041	300,584
01-January-2042 to 31-December-2042	309,868
01-January-2043 to 31-December-2043	319,152
01-January-2044 to 31-December-2044	328,436
01-January-2045 to 31-December-2045	337,720
01-January-2046 to 31-December-2046	347,004
01-January-2047 to 31-December-2047	356,288
01-January-2048 to 31-December-2048	365,572
01-January-2049 to 31-December-2049	374,856
01-January-2050 to 31-December-2050	384,140
01-January-2051 to 31-December-2051	393,424
01-January-2052 to 31-December-2052	402,708
01-January-2053 to 31-July-2053	240,328
Total estimated ERRs during the first or fixed crediting period	8,111,395
Total number of years	30
Average annual ERRs	270,379.83

1.12 Description of the Project Activity

This is an AFOLU project, the main goal of which is to avoid emissions from deforestation and degradation (REDD+), in addition to wetlands rewetting and conservation (WRC). The project is not located within a jurisdiction covered by a jurisdictional REDD+ program.

Through the SBIA workshop, the local stakeholders have designed and agreed on the following key project activities to be implemented in order to address the above focal issues. The formation of project activities also integrates customary knowledge where relevant and align with village development priorities. The project activities are listed and described in the following, which are:

1. **Sustainable fishing management**: The project will improve existing and utilize suitable lands for the creation and improvement of *beje* ponds and *keramba* in the Project Zone by integrating the use of appropriate technologies and procedures for the communities to improve and maintain fish yields. In addition, the project will facilitate the marketing of fisheries along with exploring other possible income-generating options to improve the communities' livelihood throughout the project lifetime. This activity will specifically address focal issue 3.
2. **Improved waste management and sanitation facilities**: The carbon revenue of the project will be allocated for the construction of proper water filtration technologies such as bore wells and filtration systems, installation of sanitation facilities such as proper toilets, and placement of recycling and waste bins in easy-to-see designated places within the villages. This waste will then be scheduled periodically (monthly) to be picked up by boats or barges and processed into the town in coordination with the local Environmental Services Office. In tandem, consultation activities will also be conducted to educate the communities to increase their awareness and understanding regarding the importance of waste management – possibly collaborating with local educational institutions or NGOs. This activity will specifically address focal issue 3.
3. **Increasing education opportunities and facilities**: The project has already developed a carbon school and reading club for children up to middle school and the tutors, which is implemented during weekends in both villages. The carbon school teaches community members regarding climate change and the importance of forests and the environment. With the carbon revenue, the project will provide scholarship funds for students to pursue education and build school facilities, as were urged during the FGDs – prioritizing to develop middle and high schools in Batilap village to even both communities' access to schooling and education, cutting the expenses for fuel needs. The schools will also prioritize locals as teachers. This activity is related to focal issue 3.
4. **Improved electricity and healthcare**: The project will provide access to electricity and lighting for households who have no access to electricity. In addition, the project will explore renewable energy possibilities if deemed suitable and necessary to be used, such as solar panels for houses. This activity will specifically address focal issue 3.
5. **Forest and biodiversity protection**: To mitigate biodiversity loss, the project will investigate and monitor any possible illegal activities in the Project Area including places outside the Project Zone to monitor and identify leakages. Such activities include illegal logging, wildlife hunting

and trading. Any illegal cases will be confiscated and reported to the authorities, assuring that these cases are processed according to the law. The project will coordinate with the MoF and local authorities in implementing such conducts in the Project Zone. Furthermore, community members will be involved in assessing the biodiversity in the Project Area as the forest protection program is implemented. This activity will specifically address focal issues 1 and 2.

6. **Community training and capacity building**: The project aims to empower village communities as key implementers and beneficiaries, enhancing their knowledge, skills, and participation in project oversight and implementation. This can be achieved by ensuring that community members, LPHD managers, and village governments understand their respective roles in the project cycle. Therefore, the project will implement many training activities for LPHD, village government, community groups and members by employing reliable experts according to the field of work, including vocational training, dissemination of infographics in Bakumpai language, patrolling, fire prevention, and the use of *state-of-the-art* technologies, effective organizational management, and financial literacy. This activity will specifically address all the focal issues.

7. **Alternative livelihood opportunities**: As river-based livelihoods are dominant, increasing the livelihood resilience of the communities entails diversification and improvement from available resources. There are possible job creation opportunities to divert the behaviour of deforestation and poaching, such as the development of ecotourism, NTFP processing and marketing, and other possibilities that can be explored. Along with the LPHD, these activities will be integrated strategically to develop the communities' livelihood capitals through social forestry business groups, which are open to community members in terms of decision-making and management. This activity will specifically address all the focal issues.

8. **Improved land tenure clarity and security**: Based on the household survey, there is still lack of tenure security when it comes to home ownership, in which the project will provide them with better legalities. Furthermore, this activity will address any land conflict issues in the Project Zone that may arise during the project implementation as communities outside the Project Zone may be unaware of the village forest boundaries. This activity will specifically address focal issues 1 and 3.

Presently, the communities have already implemented parts of the project activities since March 2024, such as Carbon School and Reading Club, and monthly forest patrols.

1.13 Project Location

This section is not used for a draft PD.

1.14 Conditions Prior to Project Initiation

Ecosystem type: The GBRP area mainly consists of secondary peat swamp forests, interspersed with drainage canals, followed by a portion of shrublands and swamps – providing specialized habitat for unique wildlife and plant species, such as the Bornean orangutan and Ramin trees.

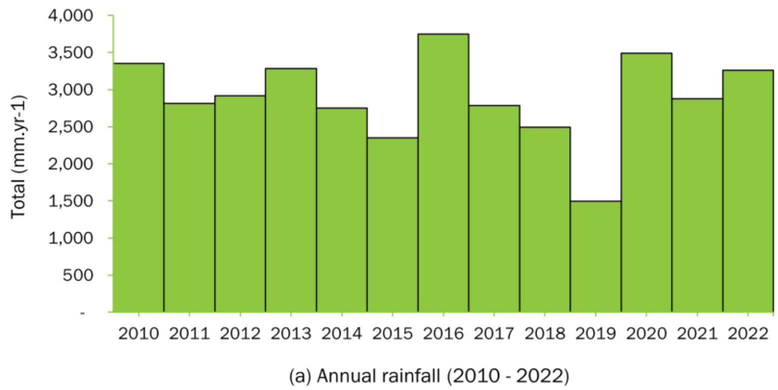
Current and historical land-use: The region had been subject to unmonitored deforestation and degradation activities, such as the construction of artificial canals for the locals to gain easy access into the forest, wildlife hunting such as birds, and intensive logging due to the existence of high-valued timber species, such as Ramin. Even after the change of status to protected forests, there are unmonitored harvesting activities of timber and non-timber forest products conducted by communities in the Project Area, which can still harm the integrity of the peat forest ecosystem. Despite having the status of the land as Village Forest where the resources are to be utilized for the benefit of the communities, inadequate protection and regulation will result in harvesting of timber and hunting of wildlife throughout the Project Area and its surroundings.

Present and prior environmental conditions of the Project Area:

The GBRP Project Area is located in Dusun Hilir Sub-District of South Barito District, among the lowland areas of Central Kalimantan Province. Most of these areas are relatively flat in slope, including Batampang and Batilap Village Forests with 0 to 8% of slant and at a relatively even altitude at 0 – 20 m asl. Meanwhile, canal topography measurements revealed a high elevation gradient in both Batampang and Batilap Village Forests.

In terms of soil formation, the soils are dominated by undecomposed organic soil, accumulating from dead plant litters that are not completely decayed due to its anaerobic environment, a core characteristic of peat soils. Meanwhile, the soil types are mostly comprised of alluvial and organosol soils. According to BPDASHL Barito (Watershed and Protected Forest Management Agency) mapping data in 2013, parts of the Project Area are categorized as critical land for rehabilitation, especially in Batilap Village Forest, while most of the area in both village forests are categorized as potentially critical land. Critical land itself is defined as lands that are no longer functioning properly due to the degradation of its physical, chemical, or biological aspect.

According to Schmidt-Ferguson climate classification based on monthly rainfall data, South Barito District falls into climate type C (slightly wet for tropical rainforest). On average, annual rainfall in the region is generally high and uneven throughout the year, which presumably has been the main cause of flooding in the two villages. In general, rainfall in South Barito Regency during the period 2010 – 2022, ranged from 2,349 – 3,490 mm per year (Figure 1.2). This gives an idea that almost all peat swamp forests in the area are almost always flooded, especially in the wet months (November – March), reinforcing the ecological uniqueness and hydrological sensitivity of the landscape.



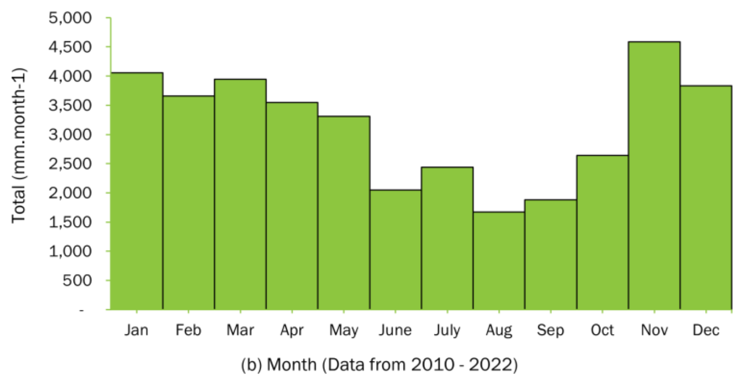


Figure 1.2 Rainfall pattern in South Barito District (2010 – 2022). (a) Annual rainfall data; (b) Monthly rainfall data disaggregated.

Barito River is the main river flowing across Gerbang Barito FMU area, the largest river that passes through 15 sub-districts from Central to South Kalimantan in the Barito Watershed. Many community members are using this river to transport logistics from Buntok Town to the villages. Both villages of Batampang and Batilap are located on the Puning River, one of the tributaries of the Barito River. Both villages rely on water from the nearby river and its surrounding water bodies for their daily livelihood activities, which is indirectly connected with the surrounding peatland forests and shrublands to regulate the quality of these waters.

Within the Barito Watershed, the village forest areas of Batampang and Batilap are hydraulically connected to the Keranen River, a sub-tributary river which flows out to the larger Puning River that passes through the two villages. Keranen River has two sub-rivers that branch depending on the route chosen to the two village forests, which community members call Keranen Lama (to Batampang Village Forest) and Keranen Baru (to Batilap Village Forest). These sub-rivers directly border the village forests' boundaries. The community members go to the forests through these river channels using their automated wooden boats.

However, the natural water balance and drainage pattern of the Project Area have been significantly affected by a network of drainage canals within the village forests, accelerating peat decomposition and carbon emissions. These canals were primarily built to facilitate timber transport and agricultural expansion before the status change of the forests to Protected Forest. These artificial drainage canals have cut the peat domes into smaller domes, which leads to uncontrolled and rapid water drainage.

In the Project Area, there are a total of 11 active drainage canals consisting of primary canals, with five canals in Batampang and six in Batilap Village Forest. In total, these canals exceed 26 kilometres in length, taking an estimated peatland area of 800 hectares. The total mapped canal length is 26.6 km, with an average length of 2.7 km in Batilap Village Forest and 2.1 km in Batampang Village Forest. Canal length distribution highlights the longest canal to be 6 km in Batilap, while the shortest is measured at 0.4 km. They accelerate belowground water flow due to the increased in-water discharge from the peatland system and lower the average water table depth to become 12.5 cm lower, with fluctuations ranging from 1 to 40 cm in depth below peat surface. There are also a few secondary canals branching off from primary canals, serving as drainage pathways. Most canals drain into Keranen River, while two canals in Batilap Village Forest flow into Puning River channels. Overall, the canal sizes vary, with widths averaging 1.8 meters in length and differing significantly between sites. Figure 1.3 below shows the canal distribution in the village forests.

Drainage canals continuously facilitate water loss from peatlands, particularly during dry months when water table depth falls below the soil surface. Without proper intervention, these drainage canals will result in peat subsidence through the reduction of peat water level. Peat subsidence exposes organic materials to oxidation, which would lead to higher CO_2 emissions. Therefore, this will result in long-term peat carbon loss, as the alteration gradually depletes the peat carbon stock overtime. Furthermore, the presence of these canals exacerbates the risks of fire due to the accelerated water loss during dry seasons. The drainage canals have increased flooding occurrences during rainy days and quickness of drying during dry seasons – the two villages have been subject to occasional flooding from heavy rainfall, with 4 flooding events recorded in 2020.

In terms of peat depth, the peat in Batampang and Batilap Village Forests are predominantly classified as deep peat (3-5 meters), with some zones containing shallower peat (1-3 meters) according to Indonesian Government maps. However, field observations and measurements conducted in April 2023 revealed that peat depth can exceed 9 meters, covering the entirety of both village forests. A total of 28 sites were surveyed for peat soil sampling, with 11 sites in Batilap and 17 Batampang, which is shown in Table 1.1 below.

Table 1.1 Peat depth classes and their area in the Project Area.

No.	Peat Depth	Area (Ha)
1	0 – 50 cm	102.58
2	50 – 300 cm	126.52
3	300 – 500 cm	246.59
4	500 – 600 cm	433.41

5	600 – 700 cm	506.06
6	700 – 800 cm	681.47
7	800 – 900 cm	7,872.19
8	900 – 1000 cm	7,220.92
9	1000 – 1200 cm	2,148.45

The following are key highlights of the peat depth mapping results:

- Shallow peat (0-300 cm): these areas, covering 229.1 hectares, are likely to be transitional zones between mineral soils and deeper peat deposits. They may indicate locations of past disturbances or natural peat formation gradients.
- Moderate peat depth (300-700 cm): found in 1,186.1 hectares, these depths are typical of secondary peat swamp forests and are essential for carbon storage and water retention.
- Deep peat (700-900 cm): this depth class dominates the landscape, covering 7,872.19 hectares, suggesting that the region has well-developed and intact peat swamp formations.
- Very deep peat (>900 cm): peat depth exceeding 900 cm cover approximately 9,600 hectares, confirming that the area holds significant long-term carbon stocks.

The peat depth mapping in both village forests revealed a wide range of peat depths, from shallow layers (0-50 cm) to exceptionally deep deposits exceeding 12 meters. According to field measurement and spatial interpolation, the majority of the Project Area is dominated by peat depths between 800 cm and 1000 cm, covering a combined area of approximately 15,000 hectares. Figure 1.3 below depicts the peat depth distribution in the Project Area.

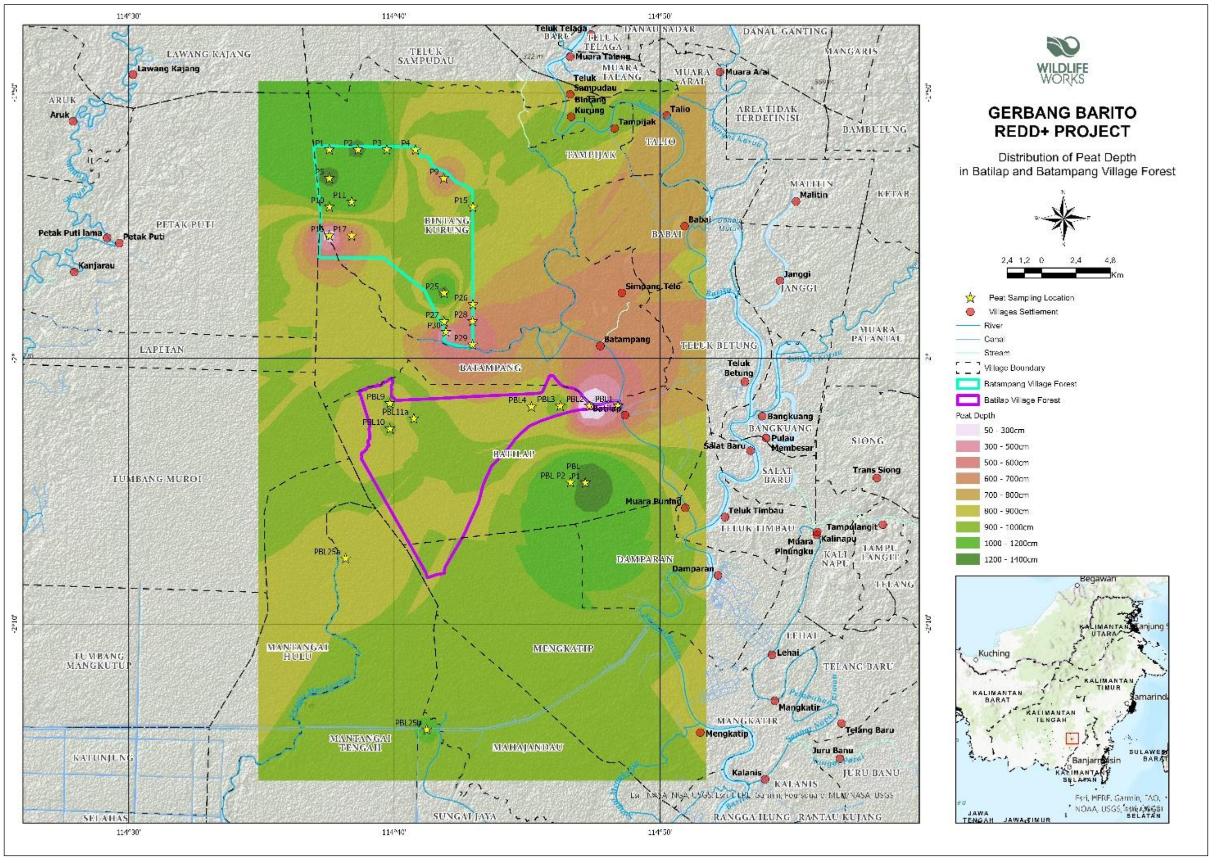


Figure 1.3 Peat depth distribution in Batampang and Batilap Village Forests.

The village forests mostly consist of a mix of intact and disturbed secondary peat swamp forests, interspersed with drainage canals, followed by shrubland, and swamps, with a tiny portion of the area being open land in Batilap Village Forest. Peat forests are naturally dense – therefore, secondary peat swamp forests are more vulnerable to disturbance, such as strong wind currents that can easily fall trees. The vegetation in the area mostly comprises of species that can adapt to the environment of the acidic and watery bodies of the peat swamp forest, including high value timber species such as Ramin (*Gonystylus bancanus* – Critically Endangered), Gemor (*Nothaphoebe coriacea*), and Belangiran (*Shorea balangeran* – Vulnerable).

1.15 Compliance with Laws, Statutes and Other Regulatory Frameworks

The following is a list of laws and regulations relevant to the implementation of REDD+ carbon projects in a village forest and the project activities in Indonesia:

- **Law No. 6/2014 on Village Government** establishes the legal framework for the recognition, governance, and empowerment of villages (*desa*) in Indonesia. This law stipulates that villages must prioritize social justice and environmental sustainability, use of local knowledge and labor, and customary land and resource rights. This law serves as the basis for supporting village-based natural resource management, including forestry, agriculture, and local enterprises. This law also allows the use of village funds to develop programs for vulnerable or informal working community members.

- **Minister of Environment and Forestry (MoEF) Regulation No. 70/2017 on Procedures for Implementing Reducing Emissions from Deforestation and Forest Degradation, Role of Conservation, Sustainable Management of Forest and Enhancement of Forest Carbon Stocks** provides the official framework and guidance for implementing REDD+ activities in Indonesia, aligning with international climate commitments and national forest governance. It operationalizes REDD+ as a measurable, reportable, and verifiable (MRV) mechanism within Indonesia's forestry and climate strategy. This regulation applies to government, private sector, and communities involved in REDD+ at the project or jurisdictional level.

  This regulation also stipulates that REDD+ initiatives and transactions must be registered through the National Registry System for Climate Change known as SRN PPI (*Sistem Registri Nasional Pengendalian Perubahan Iklim*) to avoid double counting and ensure transparency. SRN PPI (https://srn.menlhk.go.id/) is a management system that provides data and web-based information about actions and resources for Climate Change Mitigation, Climate Change Adaptation and Carbon Economic Value (*Nilai Ekonomi Karbon* or NEK) in Indonesia. It operates under the Directorate General of Climate Change Control and Carbon Economic Value Governance, Ministry of Environment. All climate-based project activities in Indonesia must be registered and recorded in this system, including validation and verification processes.

- **Law No. 7/2021 on Tax Regulation Harmonization** establishes Indonesia's carbon tax framework, setting a floor price of IDR 30 (~USD 0.0020) per kilogram of CO_2 equivalent. It serves as the foundation for future carbon pricing mechanisms, ensuring that carbon emissions are properly accounted for within Indonesia's economic and environmental policies.

- **Minister of Environment and Forestry (MoEF) Regulation No. 9/2021 on Social Forestry** provides the legal framework and technical guidance for managing *Perhutanan Sosial* (Social Forestry), ensuring community access to forest areas while promoting sustainability, legal certainty, and livelihood improvement. The regulation formalizes five schemes under the social forestry program, including *Hutan Desa* or Village Forest. Each scheme targets different community contexts but shares the goal of empowering local management of forest resources. For Village Forest, the implementing institution is LPHD, which must be formally established and recognized by the village government.

- **Government Regulation No. 23/2021 on the Administration of Forestry** governs the entire forestry sector in Indonesia, restructuring the legal framework in line with the Omnibus Law that was first imposed in 2020. It replaces and simplifies earlier regulations, aiming to promote investment, community access, environmental protection, and sustainable forest management. This regulation also supports community-based and environmental uses and enables carbon projects and ecosystem services within a national legal framework.

- **Minister of Environment and Forestry (MoEF) Regulation No. 21/2022 on Procedures for the Implementation of Carbon Economic Value** is a crucial regulation as it is aimed at providing detailed guidance on cross-sectoral carbon management, carbon trading policy through domestic or foreign trade, as well as guidelines for results-based payment; monitoring, reporting, and verification (MRV); and certification issuance under the SRN PPI. This regulation also states that carbon emission reduction projects must be registered under the national registry or SRN PPI.

- **Law No. 6/2023 in lieu of Law No. 11/2020 on Job Creation**, commonly known as Omnibus Law, overhauls over 80 regulations. After the enactment of Omnibus Law, all permits concerning Work Permits for Utilization of both Protected Forests and Production Forests regarding Area Utilization, Environmental Services, and Utilization or Collection of Timber and Non-Timber Forest Products are to be obtained using Business Permit from the Central Government.

- **Law No. 4/2023 on Development and Strengthening of Financial Sector,** often considered an Omnibus Law for the economic sector, this law formally mandates the Financial Services Authority (OJK) to regulate carbon as a financial instrument. It classifies carbon trading through a carbon exchange as a financial transaction within Indonesia's capital market framework, providing legal certainty for carbon market operations and establishing a structured financial system for domestic carbon trading.

- **Minister of Environment and Forestry (MoEF) Regulation No. 07/2023 on Procedures for Forest Carbon Trading** specifically governs carbon trading in the forestry sector and recognizes REDD+ activities, such as deforestation and degradation reduction, as eligible mechanisms for generating carbon credits. Additionally, it stipulates the type of area eligible for carbon quotas for international carbon trading, including privately owned forest concessions, areas granted to communities under social forestry scheme, and conservation area.

- **Indonesian Financial Services Authority (*Otoritas Jasa Keuangan* or OJK) Regulation No. 14/2023 on Carbon Trading through Carbon Exchange** regulates the implementation of Indonesia's carbon exchange. The regulation covers some key aspects, including (1) the requirements of the carbon exchange operator; (2) carbon market operator – operational and internal control, licensing requirements and procedures, annual work plans, budgets and reports; (3) supervision of the carbon exchange; and (4) sanctions. Subsequent to the issuance of this regulation, the national carbon trading/exchange platform (IDX Carbon) was officially launched on 26 September 2023 and now operates under OJK's supervision.

- **Minister of Environment and Forestry (MoEF) Regulation No. 12/2024 on Procedures for Carbon Trading through the Carbon Economic Value Mechanism** specifies carbon trading via the national carbon exchange platform under the Carbon Economic Value framework as mandated by Presidential Regulation No. 98/2021, which is coordinated with the Financial Services Authority (OJK) and IDX Carbon. This includes VCUs to be listed and traded on the exchange platform, with prior registration in SRN-PPI and meet technical approval requirements. The credits can be designated for domestic use, international offsetting, or Results-Based Payments depending on their classification.

- **Presidential Regulation No. 110/2025 on Implementation of Carbon Economic Value Instruments and National Greenhouse Gas Emission Control,** superseding Presidential Regulation No. 98/2021 on Carbon Economic Value, mandates the implementation and coordination of various types of carbon dioxide emission reduction mechanism, i.e., emission reduction certificate issuance, project registration under the national carbon registry (SRUK), etc. This regulation serves as the basis for the upcoming complementary ministerial regulations and technical guidelines to manage the carbon emissions reduction ecosystem in Indonesia, including the forestry sector.

The GBRP adheres to the above regulations as relevant national and local government stakeholders have been continuously being consulted with and engaged by the project team during the project development. The project team ensures that every initiative and phases in the project are conducted according to national and regional regulations.

In undertaking the utilization and business development of their village forests, the LPHDs compose two key documents: RKT (*Rencana Kerja Tahunan* or annual work plan) pertaining to yearly village forest management activities; and RKPS (*Rencana Kelola Perhutanan Sosial* or Social Forestry Management Plan) comprising a 10-year village forest management plan from 2022 to 2031. Based on these key documents, the performance of LPHD as an organization is evaluated regularly by Gerbang Barito FMU and the regional office for Social Forestry in Central Kalimantan, as stipulated in the MoEF Regulation No. 9/2021 on Social Forestry. This includes the annual reporting of activities, financial expenditures, and achievements.

1.16 Double Counting and Participation under Other GHG Programs

1.16.1 No Double Issuance

Is the project receiving or seeking credit for reductions and removals from a project activity under another GHG program?

☐ Yes ☒ No

1.16.2 Registration in Other GHG Programs

Has the project registered under any other GHG programs?

☐ Yes ☒ No

Is the project active under the other program?

☐ Yes ☐ No

1.16.3 Projects Rejected by Other GHG Programs

Has the project been rejected by any other GHG programs?

☐ Yes ☒ No

1.17 Double Claiming, Other Forms of Credit, and Scope 3 Emissions

1.17.1 No Double Claiming with Emissions Trading Programs or Binding Emission Limits

Are project reductions and removals or project activities also included in an emissions trading program or binding emission limit? See the *VCS Program Definitions* for definitions of emissions trading program and binding emission limit.

☐ Yes ☒ No

If yes, provide all required evidence of no double claiming as outlined by the VCS Standard.

1.17.2 No Double Claiming with Other Forms of Environmental Credit

Has the project activity sought, received, or is planning to receive credit from another GHG-related environmental credit system? See the *VCS Program Definitions* for definition of GHG-related environmental credit system.

☐ Yes ☒ No

If yes, provide all required evidence of no double claiming as outlined by the VCS Standard.

1.17.3 Supply Chain (Scope 3) Emissions

Do the project activities specified in Section 1.12 affect the emissions footprint of any product(s) (goods or services) that are part of a supply chain?

☐ Yes ☒ No

If yes:

Is the project proponent(s) or authorized representative a buyer or seller of the product(s) (goods or services) that are part of a supply chain?

☐ Yes ☐ No

If yes:

Has the project proponent(s) or authorized representative posted a public statement on their website saying, “Carbon credits may be issued through Verified Carbon Standard project [project ID] for the greenhouse gas emission reductions or removals associated with [project proponent or authorized representative organization name(s)] [name of product(s) whose emissions footprint is changed by the project activities]."

☐ Yes ☐ No

1.18 Sustainable Development Contributions

Indonesia's adaptation of the global Sustainable Development Goals to the village level (SDGs Desa) is regulated under Minister of Villages, Development of Disadvantaged Regions, and Transmigration Regulation (former) No. 21 of 2020, in which the SDGs Desa consist of 18 goals, closely aligned with the global 17 SDGs. In the context of the project, almost all of the project activities fall under the 18 goals, but for clarity, the project will focus on nine goals, which are:

1. **Villages without poverty (Goal 1)** – This goal is closely related to the improvement of the communities' livelihood capital. The project will prioritize the livelihood improvement of poor households, with emphasis and proper consideration of female-headed households.
2. **Quality village education (Goal 4)** – This goal is closely related with the increase of students' enrollment, pursuing higher education, and increase of school facilities, which will support early childhood development, and reducing disparities in access to schooling.
3. **Villages with clean water and sanitation (Goal 6)** – This goal is closely related with the villages' water quality, hygiene, and sanitation facilities, which are critical for the communities' quality of health and productivity.
4. **Villages with decent work and economic growth (Goal 8)** – This goal is closely related with the provision of livelihood opportunities by strengthening the rural economy by encouraging entrepreneurship, expanding employment opportunities, and promoting fair labor conditions to improve livelihood conditions. The project activities will support existing fisheries and providing other income-generating activities, working closely with the local stakeholders and poor households, with the aim of eliminating any deforestation and degradation of the forests committed by the communities.
5. **Villages without inequality (Goal 10)** – This goal ensures that marginalized and vulnerable communities have equitable access to project benefits, decision-making processes, and livelihood opportunities. It promotes social inclusion and reduces economic disparities through fair benefit-sharing and capacity-building initiatives.
6. **Environmentally conscious consumption and production in villages (Goal 12)** – This goal focuses on promoting waste reduction and management, and sustainable local and natural resource use in daily activities.
7. **Villages that are responsive to climate change (Goal 13)** – This goal emphasizes the importance of building climate resilience, disaster preparedness, and community adaptation strategies in the face of increasing environmental risks.

8. **Villages with land ecosystems that are sustainable (Goal 15)** – This goal works to preserve forests, biodiversity, and land health, promoting conservation, and sustainable fisheries and agriculture. Activities such as patrols and forest protection are closely related to this goal.

9. **Peaceful, just, and inclusive villages (Goal 16)** – This goal is closely related to the capacity building of people and organization in the villages in order to support strong village governance, dispute resolution mechanisms, and community participation to ensure social harmony and accountability. This includes LPHD as both the community-led organization and project proponent, and the village governments.

1.19 Additional Information Relevant to the Project

These sections are not used for the draft PD.

1.19.1 Leakage Management

1.19.2 Commercially Sensitive Information

1.19.3 Further Information

2 SAFEGUARDS AND STAKEHOLDER ENGAGEMENT

These sections are not used for the draft PD.

2.1 Stakeholder Engagement and Consultation

2.1.1 Stakeholder Identification

Stakeholder Identification	
Legal or customary tenure/access rights	
Stakeholder diversity and changes over time	

Expected changes in well-being	
Location of stakeholders	
Location of resources	

2.1.2 Stakeholder Consultation and Ongoing Communication

Date of stakeholder consultation	
Stakeholder engagement process	
Consultation outcome	
Ongoing communication	
Stakeholder input	

2.1.3 Free Prior and Informed Consent

Obtaining consent	
Outcome of FPIC	

2.1.4 Grievance Redress Procedure

Development process	
Grievance redress procedure	

2.1.5 Public Comments

Comments received	Actions taken
...	....

2.2 Risks to Stakeholders and the Environment

2.2.1 Management Experience

2.2.2 Risk Assessment

	Risks identified	Mitigation or preventative measure(s) taken
Natural and human-induced risks to stakeholders' wellbeing		
Risks to stakeholder participation		
Working conditions		
Safety of women and girls		
Safety of minority and marginalized groups, including children		
Pollutants (air, noise, discharges to water, generation of waste, and release of hazardous materials and chemical pesticides and fertilizers)		

2.3 Respect for Human Rights and Equity

2.3.1 Labor and Work

	Risks identified[2]	Mitigation or preventative measure(s) taken
Discrimination		
Sexual harassment		
Equal pay for equal work		
Gender equity in labor and work		
Forced labor		
Child labor		
Human trafficking		

2.3.2 Human Rights

Risks identified	Mitigation or preventative measure(s) taken

2.3.3 Indigenous Peoples and Cultural Heritage

Risks identified	Mitigation(s) or preventative measure taken

2.3.4 Property Rights

Risks identified	Mitigation or preventative measure(s) taken

2.3.5 Benefit Sharing

Process used to design the benefit sharing plan	

[2] The identified risks and commensurate mitigation or preventative measure(s) for forced labor, child labor, and human trafficking, must be inclusive of staff and contracted workers employed by third parties.

Summary of the benefit sharing plan	
Approval and dissemination of benefit sharing plan	.

2.4 Ecosystem Health

	Risks identified	Mitigation or preventative measure(s) taken
Impacts on biodiversity and ecosystems		
Soil degradation and soil erosion		
Water consumption and stress		

2.4.1 Rare, Threatened, and Endangered Species

Is the project located in or adjacent to habitats for rare, threatened, or endangered species?

☐ Yes ☐ No

Species and habitat	
Areas needed for habitat connectivity	
…	…

	Risks identified	Mitigation or preventative measure(s) taken

Habitats for rare, threatened, and endangered species		
Areas for habitat connectivity		

2.4.2 Introduction of Species

Species introduced	Classification	Justification for use	Adverse effects and mitigation

Existing invasive species	Mitigation measures to prevent the spread or continued existence of invasive species

	Risks identified	Mitigation or preventative measure(s) taken
Invasive species		

2.4.3 Ecosystem Conversion

	Risks identified	Mitigation or preventative measure(s) taken
Ecosystem conversion		

3 APPLICATION OF METHODOLOGY

3.1 Title and Reference of Methodology

Type (methodology, tool or module).	Reference ID, if applicable	Title	Version
Methodology	VM0007	VM0007 REDD+ Methodology Framework (REDD+MF),	1.8
Module (Carbon pool)	VMD0001	Estimation of carbon stocks in the above- and belowground biomass in live tree and non-tree pools (CP-AB)	1.2
Module (Carbon pool)	VMD0002	Estimation of carbon stocks in the dead-wood pool (CP-D)	1.1
Module (Carbon pool)	VMD0004	Estimation of carbon stocks in the soil organic carbon pool (CP-S)	1.1
Module (Baseline)	VMD0007	Estimation of baseline carbon stock changes and greenhouse gas emissions from unplanned deforestation and unplanned wetland degradation (BL-UP)	3.3
Module (Baseline)	VMD0042	Estimation of baseline soil carbon stock changes and greenhouse gas emissions in peatland rewetting and conservation project activities (BL-PEAT)	1.1
Module (Baseline)	VMD0016	Methods for stratification of the PA (X-STR)	1.3
Module (Baseline)	VMD0017	Methods for determining estimates of uncertainty in REDD and WRC activities (X-UNC)	2.2
Module (Emissions)	VMD0010	Estimation of emission from activity shifting for avoiding unplanned deforestation and avoiding unplanned wetland degradation (LK-ASU)	1.2
Module (Emissions)	VMD0013	Estimation of greenhouse gas emissions from biomass and peat burning (E-BPB)	1.3
Module (Emissions)	VMD0044	Estimation of emissions from ecological leakage (LK-ECO)	1.1

Module (Monitoring)	VMD0015	Methods for monitoring GHG emissions and removals in REDD and CIW projects (M-REDD)	2.3
Module (Monitoring)	VMD0046	Methods for monitoring soil carbon stock changes and greenhouse gas emissions and removals in peatland rewetting and conservation project activities (M-PEAT)	1.1
Tool	VT0001	Additionality	
Tool	T-SIG	Tool for testing the significance of GHG emissions in A/R CDM project activities	01

3.2 Applicability of Methodology

Methodology ID	Applicability condition	Justification of compliance
VM0007 – REDD-MF – Section 4.2.1 – All REDD Activity Types	Land in the Project Area has qualified as forest (following the definition used by VCS) for at least 10 years prior to project start date.	Condition met. The PA meets the qualification of forest (secondary forest) as defined by Indonesia's FREL submission in 2022, i.e., a land area of more than 6.25 ha with trees higher than 5 meters at maturity and a canopy cover of more than 30 percent. Since 2012, there has been no significant clearing or conversion of the PA. This evidence is presented in Section **Error! Reference source not found.**.
VM0007 – REDD-MF – Section 4.2.1 – All REDD Activity Types	Where land within the project area is peatland or tidal wetlands and emissions from the SOC pool are deemed significant, the relevant WRC modules are applied	Condition met. The PA is categorized as secondary peat swamp forest (peatland) according to VCS definition. WRC modules are applied in this project.

	alongside other relevant modules.	Peatland: An area with a layer of naturally accumulated organic material (peat) that meets an internationally accepted threshold (e.g., host-country, FAO, or IPCC) for the depth of the peat layer and the percentage of organic material composition. Peat originates because of water saturation. Peat soil is either saturated with water for long periods or artificially drained.
VM0007 – REDD-MF – Section 4.2.1 – All REDD Activity Types	Baseline deforestation in the project area falls within either of the following categories: 1. Unplanned deforestation (VCS category AUDef) 2. Planned deforestation/degradation (VCS category APDef)	Condition met. Baseline drivers of deforestation and degradation show unplanned deforestation activities such as timber harvesting in several areas. Unplanned deforestation: Unplanned or unsanctioned deforestation generally occurs as a result of socioeconomic forces that promote alternative uses of forested land, and the inability of institutions to control these activities. Examples include population growth, road expansion, and other infrastructure developments often lead to subsistence food production and fuel wood gathering taking place on lands not designated for such activities.

VM0007 – REDD-MF – Section 4.2.1 – All REDD Activity Types	Leakage prevention activities include: 1. Flooding agricultural lands to increase production (e.g., rice paddies); and/or Intensifying livestock production through use of feed-lots and/or manure lagoons	Condition met. The project does not include flooded agricultural lands and intensified livestock production areas to avoid leakage.
VM0007 – REDD-MF – Section 4.2.2 – AUD	Avoiding Unplanned Deforestation activities are applicable under the following conditions: Baseline agents of deforestation meet all of the following criteria: 1. Clear the land for tree harvesting, settlements, crop production (agriculturalist), ranching or aquaculture, where such clearing for crop production, ranching or aquaculture does not amount to large scale industrial agriculture or aquaculture activities; 2. Have no documented and uncontested legal right to deforest the land for these purposes; and 3. Are either residents in the reference region for deforestation or immigrants.	Condition met. Current deforestation activities mostly occur due to settlement needs (housing) of communities. With the status of PA as village forest (protected forest), timber harvesting is prohibited, while NTFP harvesting is allowed. Harvesters are either residents from the beneficiary villages or other nearby villages within the Project Zone.

	Under any other condition this methodology must not be used.	
VM0007 – REDD-MF – Section 4.2.2 – AUD	Avoiding Unplanned Deforestation activities are not applicable under the following condition: Post-deforestation land use in the baseline scenario constitutes reforestation.	Condition met. The current land use in the PA does not include reforestation activities.
VM0007 – REDD-MF – Section 4.3.1 – All WRC Activity Types	WRC activities are not eligible under the following conditions: 1. Project activities lower the water table, unless the project converts open water to tidal wetlands, or improves the hydrological connection to impounded waters. 2. Changes in hydrology do not result in the accumulation or maintenance of SOC stock, noting that a) this pertains to projects that intend to sequester carbon through sedimentation and/or vegetation development and b) this does not pertain to projects that increase salinity to reduce CH_4 emissions. Projects that aim to decrease CH_4 emissions through increased salinity must account for	Condition met. Project activities in the PA are designed to restore and maintain the natural peat water table. The project does not allow activities that result in GHG emissions in the PA, such as the burning of organic soil and use of nitrogen fertilizers.

	any changes in SOC stocks. 3. Hydrological connectivity of the project area with adjacent areas leads to a significant increase in GHG emissions outside the project area. 4. Project activities include the burning of organic soil Nitrogen fertilizer(s), such as chemical fertilizer or manure, are applied in the project area during the project crediting period.	
VM0007 – REDD-MF – Section 4.3.2 – RWE Project Activities	RWE project activities are applicable under the following conditions: Prior to the project start date, the project area meets one of the following criteria: a) The project area is free of any land use that may be displaced outside the project area, as demonstrated by at least one of the following, where relevant: i. The project area has been abandoned for two or more years prior to the project start date; or ii. Use of the project area for	Condition c) is met. The baseline scenario of the PA projected increasing activities of deforestation, construction of drainage canals, commercial fishing, and possibly, conversion of land for agricultural uses due to community needs and climate change. Proof of documents for verification of this land use condition can be found as Annex

	commercial purposes (i.e., trade) is not profitable as a result of salinity intrusion, market forces or other factors. In addition, timber harvesting in the baseline scenario within the project area does not occur; or iii. Degradation of additional wetlands for new agricultural/aquacultural sites within the country will not occur or is prohibited by enforced law. OR b) The area is under a land use that may be displaced outside the project area, although in such case, baseline emissions from this land use must not be accounted for, and where degradation of additional wetlands for new agricultural/aquacultural sites within the country will not occur or is prohibited by enforced law. OR	

	c) The area is under land use that will continue at a similar or greater level of service of production during the project crediting period (e.g., reed or hay harvesting, collection of fuelwood, subsistence harvesting, commercial fishing). The project proponent must demonstrate (a), (b) or (c) above, based on verifiable information such as laws and bylaws, management plans, annual reports, annual accounts, market studies, government studies or land use planning reports and documents.	
VM0007 – REDD-MF – Section 4.3.2 – RWE Project Activities	Peatland Rewetting: Rewetting drained peatland (RDP) activities occur on project areas that meet the VCS definition for peatland.	Condition met. The project area is defined as peatland according to VCS definitions (peat depth, percentage of organic material). See Section **Error! Reference source not found.**.
VM0007 – REDD-MF – Section 4.3.2 – RWE Project Activities	Tidal Wetland Restoration: Project activities restoring tidal wetlands include any of the following, or a combination of the following: a) Creating, restoring, and/or managing hydrological conditions (e.g., removing tidal barriers, improving hydrological connectivity, restoring tidal flow to	Condition met. In terms of restoration, project activities will include a), b), and e).

	wetlands or lowering water levels on impounded wetlands) b) Altering sediment supply (e.g., beneficial use of dredge material or diverting river sediments to sediment-starved areas) c) Changing salinity characteristics (e.g., restoring tidal flow to tidally restricted areas) d) Improving water quality (e.g., reducing nutrient loads leading to improved water clarity to expand seagrass meadows, recovering tidal and other hydrologic flushing and exchange or reducing nutrient residence time) e) (Re-)introducing native plant communities (e.g., reseeding or replanting) f) Improving management practice(s) (e.g., removing invasive species, reduced grazing) Prescribed burning of herbaceous and shrub aboveground biomass (cover burns)	
VM0007 – REDD-MF – Section 4.3.3 – CIW Project Activities	Conservation of undrained and partially drained peatland (CUPP) activities are applicable under the following conditions:	Condition met. Baseline deforestation and degradation activities occur in PA, which is peatland as defined by VCS. Project

	1. Activities occur on project areas that meet the VCS definition of peatland. 2. Project activities conserving tidal wetlands include: a. Protecting at-risk wetlands (e.g., establishing conservation easements, establishing community supported management agreements, establishing protective government regulations, and preventing disruption of water and/or sediment supply to wetland areas) b. Improving water management on drained wetlands c. Maintaining or improving water quality for seagrass meadows d. Recharging sediment to avoid drowning of coastal wetlands Creating accommodation space for wetlands migrating with sea-level rise	activities will also include protecting wetlands, maintaining and restoring the peat water table through normalizing drainage canals, and other measures to restore the natural hydrological state of the swamp forest.

VM0007 – REDD-MF – Section 4.3.3 – CIW Project Activities	Avoiding Unplanned Wetland Degradation (AUWD): Baseline agents of wetland degradation in avoiding unplanned wetland degradation activities meet all of the following criteria: a) Cause an alteration in the hydrology of the project area (involving drainage, interrupted sediment supply or both) and/or a loss of soil organic carbon b) Have no documented and uncontested legal right to degrade the wetland, and c) Are either residents in the reference region for wetland degradation or immigrants. Under any other condition, this methodology is not applicable for avoiding unplanned wetland degradation activities.	All of the criteria are met in the baseline scenario. The PA is subject to construction of drainage canals and there are no documents legalizing the degradation of wetlands in the area. Agents of deforestation and degradation of wetlands are mostly communities who reside in the Project Zone.
VMD0001 (CP-AB)	This module is applicable to all forest types and age classes.	This module is applicable as the Project Area is a peat swamp forest. Aboveground biomass of large trees is included in the project carbon accounting as it is mandatory for REDD under VM0007.
VMD0002 (CP-D)	This module is applicable to all forest types and age classes.	Condition met.

VMD0004 (CP-S)	This module is applicable to non-organic soils under all forest types and age classes.	The Project Area is classified as peatland according to VCS definitions, containing organic matter or SOC in waterlogged conditions. SOC or soil in this case is included in the carbon pool for project accounting.
VMD0007 (BL-UP)	The module is applicable for estimating baseline emissions from unplanned deforestation (conversion of forest land to non-forest land in the baseline case. The forest landscape configuration can be mosaic, transition, or frontier. The conditions of unplanned deforestation under VM0007 must be met.	Condition met (VM0007 unplanned deforestation criteria). Baseline emission estimates do not include fuelwood collection. Therefore, inclusion of baseline estimates only includes unplanned deforestation activities.
VMD0010 (LK-ASU)	This module is applicable for estimating carbon stock changes and greenhouse gas emissions related to the displacement of activities that cause deforestation of lands or wetland degradation outside the project area due to avoiding unplanned deforestation or avoiding unplanned wetland degradation in the project area. Activities subject to potential displacement are conversion of forest land to grazing lands, crop lands, and other land uses, or conversion of intact or partially degraded	This module is mandatory as it is applied along with VMD0007. Potential displaced activities in the project match the criteria as provided in the module.

	wetlands to drained or degraded wetlands. The module is mandatory if module BL-UP has been used to define the baseline and the applicability conditions in module BL-UP must be complied with in full.	
VMD0016 (X-STR)	Mandatory module for project activities based on VM0007 – REDD-MF v1.8	Single stratum of pre-deforestation in the PA was defined according to the X-STR module. The module is used in PA land cover assessment to determine treatment of project activities, quantification of baseline emissions, and carbon stock estimates. Stratum of peat
VMD0017 (X-UNC)	Used to determine the uncertainty of emission and removal estimates of CO_2 generated from REDD and WRC activities. The X-UNC module focuses on the following sources of uncertainty: 1. Determining rates of deforestation and degradation 2. Uncertainty regarding estimates of stocks in carbon pools and changes in carbon stocks 3. Uncertainty regarding peat emissions	This module is applicable as the GBRP consists of REDD and WRC activities. This module is used to estimate the uncertainty of CO_2 emissions generated from project activities.

	Uncertainty regarding project emissions	
VMD0042 (BL-PEAT)	This module is applicable to RDP and CUPP activities on project areas that meet the VCS definition for peatland. The scope of this module is limited to domed peatlands in the tropical climate zone. The following applicability conditions apply: • It must be demonstrated by using the latest version of T-SIG that N_2O emissions in the project scenario are not significant, or that N_2O emissions will not increase in the project scenario compared to the baseline scenario, and therefore N_2O emissions need not be accounted for. In the baseline scenario, the peatland must be drained or partially drained. At the start of the project, the peatland may still be undrained.	This module is applicable as the Project Area is defined as peatland in a tropical climate region, including domed peatland, and was partially drained with the construction of several artificial drainage canals. [Insert T-SIG projection of N_2O emissions]
VMD0013 (E-BPB)	This module is applicable to REDD project activities with emissions from biomass burning and REDD-WRC project activities with emissions from biomass and/or peat burning. This module is also applicable to RWE and ARR-RWE project	The project encompasses REDD and WRC activities. Emissions from burning may occur during project implementation, which are included in the GHG emissions accounting.

	activities with emissions from peat burning.	
VMD0015 (M-REDD)	The module is mandatory for REDD, CIW-REDD, RWE-REDD and stand-alone CIW project activities.	This module is used as the project encompasses REDD and WRC (CIW and RWE) activities.
VMD0044 (LK-ECO)	This module is applicable under the following condition: Leakage caused by hydrological connectivity is avoided by project design and site selection, as set out in Section 5 (Procedure) of the module.	Through RWE activities, the project ensures that GHG emissions outside the Project Area do not result from artificial alteration of hydrological aspects or ecological leakage.
VMD0046 (M-PEAT)	This module is applicable to RDP and CUPP activities as defined in the VCS Standard. The project area must meet the VCS definition for peatland. This module is limited to domed peatlands in the tropical climate zone. The following applicability conditions apply: • It must be demonstrated using tool T-SIG that N_2O emissions in the project scenario are not significant, or it must be demonstrated that N_2O emissions will not increase in the project scenario compared to the baseline scenario, and therefore N_2O emissions need not be accounted for.	This module is applicable as the Project Area is defined as peatland in a tropical climate region, including domed peatland, and was partially drained with the construction of several artificial drainage canals. [Insert T-SIG projection of N20 emissions]

	• In the baseline scenario, the peatland must be drained or partially drained. • At project start, the peatland may still be undrained. • It must be demonstrated using module LK-ECO that ecological leakage must not occur.	
T-SIG tool	[Fill in details]	This tool is used to demonstrate that N_2O emissions in the project scenario are not significant compared to the baseline scenario and justify its omission from GHG sources inclusion in the project.
VCS AFOLU Non-Permanence Risk Tool v4.2	[Fill in details]	Mandatory tool to be used in order to determine the number of buffer credits deposited into the AFOLU pooled buffer account, based on the determined project activities.

3.3 Project Boundary

This section is not used for the draft PD.

Source		Gas	Included?	Justification/Explanation
Baseline	Source 1	CO_2		
		CH_4		
		N_2O		
		Other		
	Source 2	CO_2		

Source		Gas	Included?	Justification/Explanation
		CH_4		
		N_2O		
		Other		
Project	Source 1	CO_2		
		CH_4		
		N_2O		
		Other		
	Source 2	CO_2		
		CH_4		
		N_2O		
		Other		

3.4 Baseline Scenario

This section is not used for the draft PD.

3.5 Additionality

This section is not used for the draft PD.

3.5.1 Regulatory Surplus

Is the project located in an UNFCCC Annex 1 or Non-Annex 1 country?

☐ Annex 1 country ☐ Non-Annex 1 country

Are the project activities mandated by any law, statute, or other regulatory framework?

☐ Yes ☐ No

If the project is located inside a Non-Annex 1 country and the project activities are mandated by a law, statute, or other regulatory framework, are such laws, statutes, or regulatory frameworks systematically enforced?

☐ Yes ☐ No

3.5.2 Additionality Methods

3.6 Methodology Deviations

This section is not used for the draft PD.

4 QUANTIFICATION OF ESTIMATED GHG EMISSION REDUCTIONS AND REMOVALS

These sections are not used for the draft PD.

4.1 Baseline Emissions

4.2 Project Emissions

4.3 Leakage Emissions

4.4 Estimated GHG Emission Reductions and Carbon Dioxide Removals

5 MONITORING

These sections are not used for the draft PD.

5.1 Data and Parameters Available at Validation

5.2 Data and Parameters Monitored

5.3 Monitoring Plan

APPENDIX 1: COMMERCIALLY SENSITIVE INFORMATION

Section	Information	Justification